Physical Consistency Starts at Supervision: Evidential Actor Surface Fields for Dynamic Driving Worlds

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Abstract

Driving-world reconstruction can appear safer by removing uncertain Actors, including the legitimate hazards that matter for evaluation. We instead define physical consistency from ground-truth ray and surface supervision while keeping Actor identity, trajectory, extent, and hazard state immutable. Our compiler first separates image-rendering Gaussians from an Actor-canonical physical surface and exposes observed free, occupied, and unknown evidence without turning unknown mass into a deletion rule. On frozen Argoverse 2 data, the non-learned compiler reduces LiDAR depth error from 0.207 to 0.154 m and symmetric Chamfer from 0.254 to 0.175 m while retaining every Actor and hazard state. We then train geometry on nuScenes: each completion seed generates a small set of child surfaces under motion-corrected target-set, local-plane, scale, per-frame coverage, literal first-return, and free-before-hit losses. With no deployment filter, this model reduces hazardous early returns by 5.12%, improves Chamfer by 6.21 mm, and raises hit recall by 2.76 points on an exposed development fold. Frozen producer evidence further predicts continuous FREE/OCCUPIED/UNKNOWN authority. Interpreting that authority as a categorical surface-return measure reduces hazardous early returns by 0.62 points and raises hazardous hit recall by 2.32 points relative to unit energy; treating it as additive opacity fails. Finally, we derive an exact boundary for normalized mixtures: attenuating component j lowers the pre-hit CDF iff its component CDF exceeds the current mixture CDF, and attenuation magnitude cannot reverse this sign. A failed learned-visibility ablation raises the boundary CDF on 59.32% of 99,208 rays, matching the predicted non-monotonicity. Typed ownership and inverse-query SE(3) composition keep appearance, canonical geometry, and rigid motion structurally separate. A single frozen 20-log nuScenes-to-AV2 evaluation is pending, so we claim source-domain supervised physical completion and a safety boundary, not domain invariance or real-road safety.

1. Introduction

Reconstructed driving worlds are increasingly used for perception analysis, counterfactual replay, and planning evaluation. Their visual geometry, however, is rarely a literal physical model. A Gaussian can explain image appearance while occupying observed free space, duplicating an Actor shell, or flickering across time. Conversely, a valid nearby Actor can be uncertain precisely because it is partially observed. Deleting that Actor improves an early-collision proxy without making the world more faithful. Physical consistency must therefore be defined before inference—through targets, supervision, representation, and state ownership—rather than certified after inference by a confidence filter.

We study this problem for dynamic annotated Actors. Multi-frame LiDAR is transformed into an Actor-canonical frame and split into build evidence and disjoint target sweeps. Actor identity, trajectory, extent, lifecycle, and hazard state remain immutable. Observed anchors are also fixed; each completion seed instead predicts a small set of child surfaces trained by target-set, local-plane, scale, per-frame coverage, literal first-return, and observed free-before-hit losses. Every child remains present at deployment. Thus early-return, hit-recall, and Chamfer improvements must come from the learned physical representation, not from UNKNOWN masking or Actor deletion.

This construction exposes a second distinction: surface geometry and sensor termination are different physical variables. A well-placed Gaussian can still receive contradictory free and occupied evidence from different views. We retain that evidence at the data producer and supervise continuous FREE/OCCUPIED/UNKNOWN authority for anchors and completion children. Treating the occupied coordinate as additive opacity fails because total optical thickness depends on primitive density. We instead compose frozen geometry and learned authority into one normalized categorical surface-return measure. Its CDF is the deployed first-return distribution, so training and inference share the same object and no post-hoc threshold is needed.

Dynamic motion, static context, and appearance are kept outside this geometry–termination coupling. Rigid Actor pose relocates the canonical field through inverse SE(3) querying but cannot deform it. Static Background has a different owner. Image gradients update only a sibling visual state through a stop-gradient physical carrier. These typed interfaces do not solve pose, visibility, or appearance; they prevent those variables from silently compensating for physical geometry during optimization.

Finally, the normalized return measure permits an exact safety analysis. Attenuating one primitive does not necessarily lower probability before a hazardous boundary: the sign depends on whether that primitive's own pre-boundary CDF lies above or below the mixture CDF. We derive the finite change and show that attenuation magnitude cannot reverse its sign. A failed learned- visibility experiment then becomes an interpretable boundary rather than motivation for another filter. We verify this effect on 99,208 frozen rays and never use the target-dependent sign as a deployment gate.

Our contributions are:

- supervision-native Actor completion whose GT set, ray, and free-space losses remain active through physical training, with no UNKNOWN or confidence deletion at deployment;
- producer-evidential FREE/OCCUPIED/UNKNOWN authority represented as a categorical surface-return measure rather than density-dependent volumetric opacity;
- typed canonical-geometry, rigid-pose, static-world, and appearance ownership with inverse-query SE(3) equivariance and no image-gradient path to physical geometry;
- an exact finite attenuation boundary for normalized mixtures and a frozen nuScenes-to-AV2 protocol that withholds cross-domain claims until its complete 20-log aggregate is available.

2. Related Work

From visual reconstruction to physical returns. 3D Gaussian Splatting represents appearance efficiently [13], and driving simulators such as UniSim, LiDAR4D, and DyNFL optimize visual or LiDAR observations over time [23, 26, 30]. Gaussian opacity, however, is not automatically occupancy or collision geometry. Automotive LiDAR supplies both endpoints and measured free-space intervals [7]. Neural LiDAR Fields models first returns with a ray-weight distribution [8]; categorical depth networks learn a metric depth mode [17]; QueryOcc directly samples positive and negative continuous queries [15]. We retain their relevant boundary: native rays supervise the physical representation, while an Actor-local surface and its termination distribution remain different typed objects.

Completion and evidential occupancy. Object-centric occupancy completion canonicalizes long LiDAR tracklets before implicit decoding [29], while SelfOcc and SparseOcc++ learn continuous geometry fields [9, 20]. PoinTr translates point sets and SnowflakeNet grows child points from parent seeds [25, 27]. Gau-Occ initializes Gaussians from diffusion-completed LiDAR, but its multimodal occupancy head may still update center, scale, and rotation [16]. We borrow target/set completion, not unconstrained image-conditioned geometry: dynamic sweeps are Actor-motion compensated, observed anchors are immutable, and geometry/ray losses remain active throughout.

ALSO and evidential occupancy derive FREE/OCCUPIED supervision from the sensor origin [1, 11, 12]. GaussianFormer-2 and GaussRender connect Gaussian representations to occupancy and projective supervision [3, 10]; Vol3DGS analytically integrates Gaussians into transmittance [19]. Our negative optical ablations expose a complementary issue: additive thickness changes with primitive density. We instead use producer evidence as categorical surface-return authority and state exactly what its normalized CDF can and cannot guarantee.

Dynamic factorization and appearance ownership. Multi-frame completion labels can contain moving-object traces [24]. DualAD separates static and dynamic streams with ego/object-motion compensation [4]; DynamicVGGT predicts Gaussian velocity under scene-flow supervision [6]; DeGO separates rigid offsets and non-rigid deformation [5]; and SelfOccFlow maintains separate static/dynamic signed-distance fields [21]. These works motivate our refusal to hide motion inside a canonical shape head. We use annotated rigid pose only as a read-only transform; non-rigid evolution remains unsupported without its own target. Likewise, Feature 3DGS and Neural Shell Texture Splatting show how visual attributes can live on geometry [28, 31]. Our stricter ownership contract gives image gradients a sibling visual state and no path to physical center or scale.

Cross-sensor evaluation. nuScenes and Argoverse 2 provide calibrated but materially different multi-sensor trajectories [2, 22]. LiDAR domain shift includes sampling pattern as well as scene statistics; source-only density consistency and accumulated sequence geometry can improve transfer [14, 18]. They do not certify Actor-level physical transfer. We therefore freeze one AV2 cohort before reading any learned output and prohibit target adaptation, calibration, or failed-log deletion.

3. Problem Formulation

Let Actor i have an immutable semantic state

$$A_i = (ID_i, X_{i,0:H}, D_i), \tag{1}$$

containing identity, trajectory, and dimensions. Appearance primitives may be associated with A_i, but collision queries consume only an Actor-local physical surface S_i^{phys} and its known/unknown mask.

We separate two random variables. Artifactness a_i depends on sensor and geometric evidence E_i, physical surface state, and provenance:

$$a_i = P(\text{artifact} \mid E_i, S_i^{\text{phys}}, P_i). \tag{2}$$

Hazardness h_i depends on Actor dynamics, map context, and a candidate Ego trajectory τ:

$$h_i = P(\text{hazard} \mid X_{i,0:H}, \tau, \mathcal{M}). \tag{3}$$

The goal is paired conditional invariance, not unconditional statistical independence. For the same clean Actor, a legal safe-to-hazard intervention should preserve a_i. For the same behavior, a clean-to-artifact intervention should preserve h_i.

For task reliability, let $n_{\tau,t}$ be the local boundary normal, $d_{i,t}(\tau)$ the signed Actor clearance, and $\epsilon_{i,t}$ the Actor residual. We define

$$C(\tau, H) = \max_{i,t \le H} \frac{|n_{\tau,t}^\top \epsilon_{i,t}|}{\max(|d_{i,t}(\tau)|, \varepsilon)}. \tag{4}$$

The runtime interface returns $R(\tau, H, b) = P(C \le b)$ and enforces $\partial R/\partial b \ge 0$ and $\partial R/\partial H \le 0$.

4. Method

4.1. Canonical Evidence and Immutable Actor State

For Actor i at time t, annotated sensor, ego, and Actor poses map a LiDAR endpoint into a canonical coordinate system,

$$x_{i,t} = T_{i,t}^{-1} T_{\text{ego},t} T_{\text{sensor}} x_t^{\text{sensor}}. \tag{5}$$

Build sweeps provide model input; disjoint held-out sweeps construct the target T_i and are never input features. Directly observed and matched projected endpoints form an immutable anchor set A_i. Actor identity, trajectory, extent, lifecycle, and hazard label are immutable interfaces rather than prediction targets.

For a held-out ray $r = (o, v, d^\star)$ and physical surface S, the literal first return is

$$d_S(r) = \min\{s > 0 : o + sv \in S_\epsilon\}, \qquad \min \varnothing = \infty. \tag{6}$$

An early return satisfies $d_S(r) < d^\star - \epsilon$. Deletion can only make this predicate non-increasing, but may destroy hits and surface coverage. We therefore do not treat a lower early-return rate as sufficient evidence of consistency.

4.2. Supervision-Native Child Surfaces

One completion seed cannot cover a dense target surface. Each seed x_j predicts four bounded Actor-frame residuals and positive support scales,

$$x'_{jk} = x_j + \Delta x_{jk}(E_i), \qquad S_i^\theta = A_i \cup \{(x'_{jk}, s_{jk})\}_{j,k=1}^4, \tag{7}$$

where E_i contains build-only geometric and ray evidence. Geometry pretraining combines bidirectional set distance, target-local point-to-plane distance, and a target-spacing scale loss. Physical fine-tuning keeps these terms active and adds frame-balanced target coverage, differentiable literal first return, and observed free-before-hit supervision:

$$\mathcal{L}_{\text{phys}} = \mathcal{L}_{\text{set}} + .25\mathcal{L}_{\text{plane}} + .1\mathcal{L}_{\text{scale}} + \mathcal{L}_{\text{frame}} + \mathcal{L}_{\text{first}} + .5\mathcal{L}_{\text{free}}. \tag{8}$$

Space behind the first hit is not labeled free. Every child remains in S_i^θ at deployment; UNKNOWN is evidence state, not a mask. Thus an improvement must arise through the supervised representation rather than selective output removal.

4.3. Producer-Evidential Return Authority

Surface support and sensor termination are distinct. For each anchor or child j, the build descriptor e_j retains source ray/frame identity, construction provenance, projection displacement, canonical hit count, temporal/view support, and continuous build evidence $m_j^{\text{build}} = (m_F, m_O, m_U)$. Disjoint target rays construct m_j^{GT} from free evidence before a return, occupied evidence near it, and unknown evidence behind it or outside ray support. A bounded head predicts $m_j = f_\phi(e_j)$ and is supervised by soft three-state cross-entropy. Completion children also receive the GT free-before-hit event loss used by the deployed ray objective. Geometry is frozen during this authority stage.

Let o_j be predicted occupied mass and κ_{kj} the frozen Gaussian support of primitive j at ordered depth d_k. Instead of adding $o_j \kappa_{kj}$ as density-dependent opacity, we define one joint categorical return measure,

$$q(k,j) = \frac{o_j \kappa_{kj}}{\sum_{a,b} o_b \kappa_{ab}}, \qquad p_k = \sum_j q(k,j), \qquad F_k = \sum_{\ell \le k} p_\ell. \tag{9}$$

The deployed return is the median, $\widehat{d} = \min\{d_k : F_k \ge 1/2\}$. Training and deployment use the same ordered distribution. Normalization guarantees non-negative unit mass and a monotone CDF, but does not turn o_j into occupancy or collision probability. The current protocol conditions on one return inside the Actor box and does not model ray drop.

For boundary $b = d^\star - \epsilon$, piecewise-linear interpolation gives the exact deployed decision

$$\widehat{d} < b \iff F(b) > \tfrac{1}{2}, \qquad F(b) = F_{\text{anchor}}(b) + F_{\text{child}}(b). \tag{10}$$

This decomposition explains responsibility but is never used to delete a primitive or case.

4.4. Typed Motion and Appearance Ownership

Rigid pose is a separate read-only authority. With $T_{it} = (R_{it}, t_{it})$, the physical query is

$$Q_{it}(x; P_i, T_{it}) = \log \sum_j \exp\left[-\frac{\|R_{it}^\top (x - t_{it}) - c_{ij}\|_2^2}{2s_{ij}^2}\right], \tag{11}$$

where $P_i = \{(c_{ij}, s_{ij})\}_j$ is canonical physical state. Static Background is a separate scene component. Visual state V_i contains render geometry, SH, and opacity; image optimization receives $\mathrm{sg}(P_i)$ and updates only V_i. Consequently $\partial Q_{it}/\partial V_i = 0$, and for a common rigid transform g, inverse querying yields $Q_{gT}(gx) = Q_T(x)$. These are ownership and equivariance guarantees, not guarantees that pose or geometry is correct.

4.5. Finite Attenuation Boundary

The normalized measure also identifies when learned visibility is unsafe. For primitive j, let $D_j = \sum_k \kappa_{kj}$, $N_j = \sum_{d_k<b} \kappa_{kj}$ with boundary interpolation, $C_j = N_j/D_j$, and $r_j = o_j D_j / \sum_a o_a D_a$. Direct differentiation gives

$$\frac{\partial F(b)}{\partial \log o_j} = r_j\big(C_j - F(b)\big). \tag{12}$$

For any finite attenuation $v \in (0, 1]$ applied to the component, or to a family treated as one component,

$$F_v(b) - F(b) = \frac{(1-v)r_j\big(F(b) - C_j\big)}{1 - (1-v)r_j}. \tag{13}$$

Attenuation lowers the pre-boundary CDF iff $C_j > F(b)$; changing its magnitude cannot reverse the sign. Because b uses target depth, this is an explanation and model-design boundary, not a deployable oracle gate.

A distinct non-implication concerns the geometry surrogate. The differentiable renderer optimizes an alpha-composited expected depth, whereas Eq. 6 is a minimum over beam-supported surface. An arbitrarily small shallow component can move the minimum across b while perturbing the expectation arbitrarily little; with late residual mass, that perturbation may even move the expectation toward the target. Lower expected-depth error therefore cannot certify a safer literal first return without an explicit ordered support or CDF margin.

5. Experiments

5.1. Protocol and Claim Boundary

We train and make every representation choice on nuScenes. The geometry fold contains 66 Actors and is

Surface	Free (%) ↓	Recall (%) ↑	CD (m) ↓
Single-frame input	100.0	53.36	0.254
Nearest canonical	84.91	**62.60**	**0.168**
Ray-certified	**0.00**	59.56	0.175

Table 1. Frozen zero-shot AV2 quantitative result. Nearest-surface projection fits geometry best but violates matched-ray free space; ray-certified projection accepts a small Chamfer cost. Later held-out frames are target-only.

explicitly marked as development-exposed because its parent representation had already been inspected. Argoverse 2 Sensor validation is a single frozen zero-shot evaluation: before any learned-method output is read, we sort the 150 log UUIDs and choose every fifth log, yielding 20 quantitative logs. We prohibit AV2 fine-tuning, calibration, threshold selection, and failed-log deletion. At the time of this draft the aggregate is still incomplete, so no partial learned-transfer metric appears below.

Physical evaluation uses literal first-return error, hit recall within 0.20 m, symmetric Chamfer distance, and the same metrics stratified by a target-defined hazard label. Actor identity, trajectory, extent, and lifecycle are immutable. This prevents a method from improving an early-return score by removing a difficult Actor. Unless stated otherwise, all learned comparisons keep every Actor and every generated surface.

As an external, non-learned reference, the frozen ray-certified compiler operates on 433 AV2 Actors, 147 hazardous (Table 1). It reduces target depth error from 0.207 to 0.154 m and Chamfer from 0.254 to 0.175 m while preserving every Actor. Nearest-surface projection obtains lower Chamfer but violates matched observed-free rays, illustrating why average geometry alone is not a physical objective.

5.2. Supervision-Native Canonical Geometry

The train-only displacement oracle shows that a non-deletion Pareto direction exists, but a residual policy reproduces its ray gain mainly through an UNKNOWN action. We therefore remove the action and change the target. Motion-corrected held-out LiDAR is transformed into each Actor's canonical frame; every completion seed predicts four child centers and scales, and training keeps target-set, tangent-plane, scale, per-frame coverage, literal first-return, and free-before-hit losses active. No output is filtered at deployment.

Table 2 reports the resulting M8 model. Hazardous early returns decrease by 5.12% relative, Chamfer improves by 6.21 mm, and hit recall rises by 2.76 points. The three metrics move together because physical consistency is imposed on the target and loss, rather than obtained by deleting low-confidence surfaces. The representative Actor in Fig. 1 is

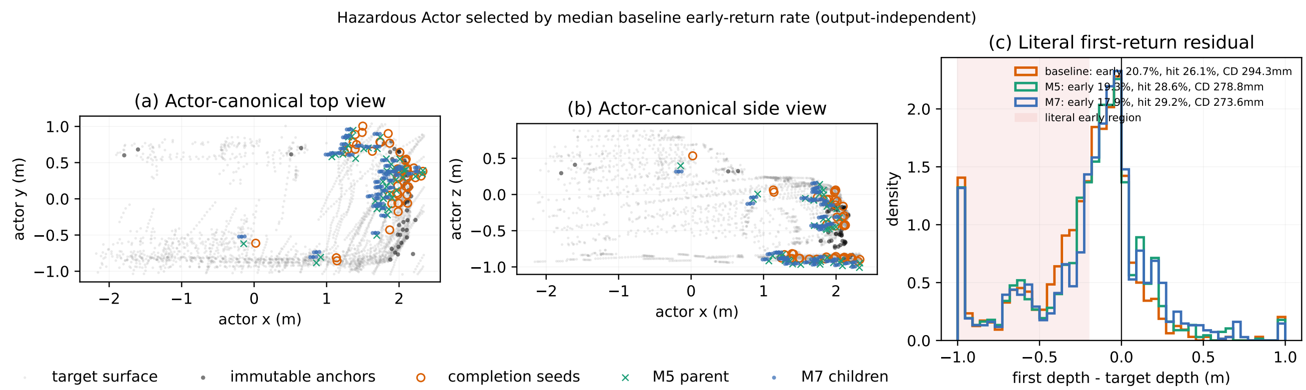

Figure 1. Supervision-native geometry on a hazardous development Actor selected without consulting the learned output. Forty-seven completion seeds expand into 188 child surfaces while observed anchors remain fixed. Baseline/M5/M7 early rates are $20.7/19.3/17.9\%$, hit rates are $26.1/28.6/29.2\%$, and Chamfer distances are $294.3/278.8/273.6$ mm. Gray held-out GT points are target-only.

Table 2. Core source-domain results on the frozen, exposed nuScenes development fold. Negative early-return and Chamfer changes are favorable; positive hit changes are favorable. Comparisons differ by panel and are stated explicitly.

Stage	Metric	Change
Geometry: M8 vs. original completion		
	hazardous early returns (relative)	-5.123%
	symmetric Chamfer	-6.207 mm
	hit recall	$+2.759$ pp
Return measure: M39 vs. unit categorical energy		
	early returns (all / hazard / clear)	$-0.52/-0.62/-0.03$ pp
	hit recall (all / hazard / clear)	$+2.17/+2.32/+1.46$ pp

selected by the *baseline* early-return median before inspecting model output.

5.3. Producer Evidence as a Categorical Return Measure

Geometry and ray termination are not the same variable. Exact producer replay over 1,004 Actors aligns 297,535 anchors with build and held-out FREE/OCCUPIED/UNKNOWN votes. Their occupied masses correlate ($r = 0.514$), but direct three-state agreement is only 43.41%; 58.72% of anchors receive both free and occupied held-out votes. Thus provenance is informative but cannot be a hard keep/delete label.

We freeze M8 geometry and learn only continuous authority from build-time producer evidence. Weighting Gaussian energy or interpreting authority as additive optical thickness fails: opacity grows with primitive density and couples endpoint termination to the front tail. The decisive M39 change leaves the learned masses untouched and interprets them as a normalized categorical surface-return measure. Relative to unit categorical energy, early returns decrease in all, hazardous, and clear strata by $0.52/0.62/0.03$ points, while hit recall rises by $2.17/2.32/1.46$ points (Table 2). These three pre-registered directions pass without changing geometry or removing a primitive. The conclusion is deliberately narrow: the mass is return evidence, not occupancy probability or collision risk.

Failure-directed variants sharpen that conclusion. Joint ray fine-tuning transfers mass to the denser child family; fixing total family mass still optimizes target-bin likelihood without controlling the deployed boundary event. Directly supervising GT not-early and hit-band events recovers aggregate and hazardous directions but regresses the clear stratum. We therefore retain frozen M39 for the single AV2 test instead of tuning another development loss.

5.4. An Exact Boundary for Learned Attenuation

M44 verifies the exact CDF accounting of Eq. 10 over 99,208 rays. M39 lowers mean pre-boundary mass by 0.331 points relative to unit energy: observed anchors contribute -0.346 points, while completion children contribute $+0.016$ points. The supported gain is therefore not a completion filter; producer-supervised anchor authority dominates it.

Local oriented support and a bounded learned visibility factor both lower their training losses but transport error between ray strata. The visibility model raises hazardous early returns by 0.483 points relative to M39. Equation 13 explains why tuning its magnitude cannot repair the sign. In Fig. 2, uniform child attenuation is analytically adverse on 58.85% of rays and safe on only 30.96%. Under the learned component-wise attenuation, adverse pressure dominates 65.45% overall and the exact pre-boundary CDF rises on 59.32% of rays. The first-order sign agrees with the finite change on 95.73% of rays. We use this identity only to explain the failure—never as an oracle gate or post-hoc

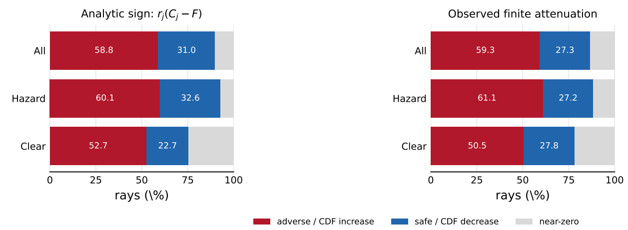

Figure 2. Exact attenuation boundary on 99,208 frozen rays. Red is analytically adverse (left) or an observed increase in pre-boundary CDF (right); blue is safe/decreasing. The diagnostic is not a deployment rule.

selector.

M51 tests the separate expected-depth/minimum boundary using M8 and rejected M50. On identical, non-voxelized support, hard early rate rises by 0.490 points; 38.03% of the newly early rays nevertheless have lower smooth absolute depth error. Under the full-anchor, voxelized deployment surface, the values are 0.545 points and 36.54%. Smooth-error and hard-depth changes have correlation 0.038, while support realization flips 1.28% of ray events. The non-implication therefore precedes deployment discretization and is not repaired by better frame weighting.

5.5. Typed Motion, Static Context, and Appearance

We audit the ownership contract separately from surface accuracy. Twelve scene-0230 Actors are identity-matched to a read-only StreetGS checkpoint. Across 36 Actor–frames, 5,791 physical Gaussians follow the checkpoint's rigid Actor poses, while 106,807 appearance-owned and 1,140,862 Background Gaussians remain outside the physical query. Inverse querying gives maximum energy and pairwise-distance residuals of 3.40×10^{-14} and 1.24×10^{-14} m under as much as 126.28 m translation. This establishes the SE(3) composition contract, not pose or geometry correctness.

Image supervision is likewise prevented from moving collision geometry. Optimizing only SH and opacity on a fixed 309-carrier Actor improves held-out footprint PSNR from 15.86 to 16.94 dB, but remains 8.41 dB below the original visual Actor. A fixed parent–surfel hierarchy improves all six held-out views to 17.73 dB, yet retains a 7.62 dB gap. The result supports typed gradient ownership and exposes insufficient appearance capacity; it does not justify sharing image-trained geometry with the physical field.

5.6. What the Evidence Establishes

The source experiments support three separable statements: GT-defined losses improve Actor-canonical geometry without deletion; producer evidence is useful when represented as categorical return authority; and normalized-mixture attenuation has a non-monotone, exactly characterizable boundary. Typed ownership keeps rigid motion, Background, appearance, and canonical geometry from silently compensating for one another. None of these statements certifies collision freedom, unobserved completion, trajectory accuracy, domain invariance, or real-road safety. The frozen AV2 aggregate will test only the learned return representation's cross-domain direction; its outcome cannot retroactively select or adapt the method.

6. Limitations

The learned results use a 66-Actor development fold whose parent representation was previously inspected. They establish a source-domain Pareto direction, not an unbiased final estimate. The single frozen 20-log AV2 aggregate is still pending; we neither inspect partial metrics nor adapt on AV2. The earlier non-learned AV2 compiler is a separate operator result and must not be presented as learned transfer.

Our categorical measure assumes one return inside an annotated Actor box. Its mass is surface-return evidence, not calibrated occupancy or collision probability; sparse sensing leaves unknown space, and neither M8 nor M39 proves watertight completion. Tracked cuboids, poses, and GT LiDAR may be noisy. Current canonicalization covers rigid annotated vehicles, while deformable Actors, static topology, explicit no-return modeling, sensor-error sets, and closed-loop vehicle dynamics remain open. Exact SE(3) equivariance certifies coordinate composition only, not trajectory or surface correctness.

The appearance audit is a single-scene mechanism study on views exposed before the final hierarchy design. It shows that image gradients can be isolated from physical geometry, but leaves a 7.62 dB gap to the original visual Actor and does not establish photorealism or cross-scene fidelity. Finally, the M49 attenuation identity is evaluated with a GT-defined boundary on exposed rays. It explains why attenuation can be adverse, but using its sign as a selector would recreate the post-hoc oracle behavior we seek to avoid. None of our results is a collision-freedom, planning, policy, domain-invariance, or real-road-safety guarantee.

7. Conclusion

Physical consistency is a property of targets, supervision, representation, and deployment together—not a score that can be repaired by deleting uncertain outputs. Motion-corrected Actor targets and persistent set/ray/free-space losses let our child-surface model improve early returns, hit recall, and Chamfer jointly on nuScenes while retaining every Actor. Producer evidence then becomes useful only when interpreted as categorical surface-return authority, not additive volumetric opacity. Our negative ablations sharpen this distinction: local orientation and learned visibility transport risk between ray strata even as their training losses fall. The exact finite attenuation boundary explains why no opacity magnitude can make this operation universally safe. Typed ownership and SE(3) inverse querying keep canonical geometry, rigid motion, static context, and appearance separate, but do not certify their accuracy. The pending frozen AV2 aggregate will determine whether the learned return representation transfers; regardless of that outcome, the source-domain supervision result and normalized-mixture safety boundary do not imply collision freedom, domain invariance, or real-road safety.

References

[1] Alexandre Boulch, Corentin Sautier, Bj"orn Michele, Gilles Puy, and Renaud Marlet. ALSO: Automotive LiDAR self-supervision by occupancy estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13455–13465, 2023. 2

[2] Holger Caesar, Varun Bankiti, Alex H. Lang, Sourabh Vora, Venice Erin Liong, Qiang Xu, Anush Krishnan, Yu Pan, Giancarlo Baldan, and Oscar Beijbom. nuScenes: A multimodal dataset for autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020. 2

[3] Loïck Chambon, Eloi Zablocki, Alexandre Boulch, Mickaël Chen, and Matthieu Cord. GaussRender: Learning 3D occupancy with gaussian rendering. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025. 2

[4] Simon Doll, Niklas Hanselmann, Lukas Schneider, Richard Schulz, Marius Cordts, Markus Enzweiler, and Hendrik P. A. Lensch. DualAD: Disentangling the dynamic and static world for end-to-end driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14728–14737, 2024. 2

[5] Yang Gao, Wuyang Li, Po-Chien Luan, and Alexandre Alahi. Deformable gaussian occupancy: Decoupling rigid and nonrigid motion with factorized distillation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 28588–28598, 2026. 2

[6] Zhuolin He, Jing Li, Guanghao Li, Xiaolei Chen, Jiacheng Tang, Siyang Zhang, Zhounan Jin, Feipeng Cai, Bin Li, Jian Pu, Jia Cai, and Xiangyang Xue. DynamicVGGT: Learning dynamic point maps for 4D scene reconstruction in autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 35670–35679, 2026. 2

[7] Peiyun Hu, Jason Ziglar, David Held, and Deva Ramanan. What you see is what you get: Exploiting visibility for 3D object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11001–11009, 2020. 2

[8] Shengyu Huang, Zan Gojcic, Zian Wang, Francis Williams, Yoni Kasten, Sanja Fidler, Konrad Schindler, and Or Litany. Neural LiDAR fields for novel view synthesis. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 18236–18246, 2023. 2

[9] Yuanhui Huang, Wenzhao Zheng, Borui Zhang, Jie Zhou, and Jiwen Lu. SelfOcc: Self-supervised vision-based 3D occupancy prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19946–19956, 2024. 2

[10] Yuanhui Huang, Amonnut Thammatadatrakoon, Wenzhao Zheng, Yunpeng Zhang, Dalong Du, and Jiwen Lu. GaussianFormer-2: Probabilistic gaussian superposition for efficient 3D occupancy prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27477–27486, 2025. 2

[11] Jonas Kälble, Sascha Wirges, Maxim Tatarchenko, and Eddy Ilg. Accurate training data for occupancy map prediction in automated driving using evidence theory. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5281–5290, 2024. 2

[12] Jonas Kälble, Sascha Wirges, Maxim Tatarchenko, and Eddy Ilg. EvOcc: Accurate semantic occupancy for automated driving using evidence theory. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27467–27476, 2025. 2

[13] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkuehler, and George Drettakis. 3D gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42 (4), 2023. 2

[14] Hyeonseong Kim, Yoonsu Kang, Changgyoon Oh, and Kuk-Jin Yoon. Single domain generalization for LiDAR semantic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17587–17598, 2023. 2

[15] Adam Lilja, Ji Lan, Junsheng Fu, and Lars Hammarstrand. QueryOcc: Query-based self-supervision for 3D semantic occupancy. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 21399–21408, 2026. 2

[16] Chengxin Lv, Yihui Li, Hongyu Yang, and YunHong Wang. Gau-Occ: Geometry-completed gaussians for multi-modal 3D occupancy prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14198–14207, 2026. 2

[17] Cody Reading, Ali Harakeh, Julia Chae, and Steven L. Waslander. Categorical depth distribution network for monocular 3D object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8555–8564, 2021. 2

[18] Jules Sanchez, Jean-Emmanuel Deschaud, and François Goulette. Domain generalization of 3D semantic segmentation in autonomous driving. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 18077–18087, 2023. 2

[19] Chinmay Talegaonkar, Yash Belhe, Ravi Ramamoorthi, and Nicholas Antipa. Volumetrically consistent 3D gaussian rasterization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10953–10963, 2025. 2

[20] Pin Tang, Zhongdao Wang, Guoqing Wang, Xiangxuan Ren, and Chao Ma. SparseOcc++: Geometry-aware sparse latent representation for semantic occupancy prediction. *arXiv preprint arXiv:2607.04732*, 2026. 2

[21] Xavier Timoneda, Markus Herb, Fabian Duerr, and Daniel Goehring. SelfOccFlow: Towards end-to-end self-supervised 3D occupancy flow prediction. *arXiv preprint arXiv:2602.23894*, 2026. 2

[22] Benjamin Wilson, William Qi, Tanmay Agarwal, John Lambert, Jagjeet Singh, Siddhesh Khandelwal, Bowen Pan, Ratnesh Kumar, Andrew Hartnett, Jhony Kaesemodel Pontes, Deva Ramanan, Peter Carr, and James Hays. Argoverse 2: Next generation datasets for self-driving perception and forecasting. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021. 2

[23] Hanfeng Wu, Xingxing Zuo, Stefan Leutenegger, Or Litany, Konrad Schindler, and Shengyu Huang. Dynamic LiDAR re-simulation using compositional neural fields. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19988–19998, 2024. 2

[24] Zhaoyang Xia, Youquan Liu, Xin Li, Xinge Zhu, Yuexin Ma, Yikang Li, Yuenan Hou, and Yu Qiao. SCPNet: Semantic scene completion on point cloud. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17642–17651, 2023. 2

[25] Peng Xiang, Xin Wen, Yu-Shen Liu, Yan-Pei Cao, Pengfei Wan, Wen Zheng, and Zhizhong Han. SnowflakeNet: Point cloud completion by snowflake point deconvolution with skip-transformer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5499–5509, 2021. 2

[26] Ze Yang, Yun Chen, Jingkang Wang, Sivabalan Manivasagam, Wei-Chiu Ma, Anqi Joyce Yang, and Raquel Urtasun. UniSim: A neural closed-loop sensor simulator. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1389–1399, 2023. 2

[27] Xumin Yu, Yongming Rao, Ziyi Wang, Zuyan Liu, Jiwen Lu, and Jie Zhou. PoinTr: Diverse point cloud completion with geometry-aware transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 12498–12507, 2021. 2

[28] Xin Zhang, Anpei Chen, Jincheng Xiong, Pinxuan Dai, Yujun Shen, and Weiwei Xu. Neural shell texture splatting: More details and fewer primitives. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 25229–25238, 2025. 2

[29] Chaoda Zheng, Feng Wang, Naiyan Wang, Shuguang Cui, and Zhen Li. Towards flexible 3D perception: Object-centric occupancy completion augments 3D object detection. In *Advances in Neural Information Processing Systems*, 2024. 2

[30] Zehan Zheng, Fan Lu, Weiyi Xue, Guang Chen, and Changjun Jiang. LiDAR4D: Dynamic neural fields for novel space-time view LiDAR synthesis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5145–5154, 2024. 2

[31] Shijie Zhou, Haoran Chang, Sicheng Jiang, Zhiwen Fan, Zehao Zhu, Dejia Xu, Pradyumna Chari, Suya You, Zhangyang Wang, and Achuta Kadambi. Feature 3DGS: Supercharging 3D gaussian splatting to enable distilled feature fields. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 21676–21685, 2024. 2